Tree decomposition for the reconstruction of ancestral RNA sequences under multiple structures with base pairing and stacking constraints.

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1 INTRODUCTION

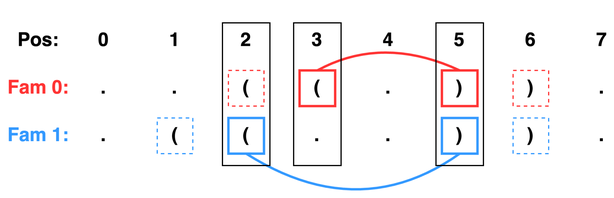
Understanding evolutionary relationships and scenarios has been one of the main keys in the development of better sequence mutation models, which culminated in the recent success of structure prediction and design methods.
While we can only observe today's environment, different models have allowed us to shed some light on how the chain of events, starting from a lone ancestor, has evolved towards the observed diversity. Validating such studies is a theoretical challenge that has brought many interesting observations and has evolved the reconstruction principles from toy models to the prediction of entire ancestral genomes[1]. The reconstruction of ancestral non-coding RNA sequences (ncRNA) is particularly challenging due to the main conservation forces being applied to the structure, rather than the sequence. Naively trying to preserve the structure during the reconstruction tends to produce sequences that are at the center of the evolutionary neutral network (sequences with the same phenotype). This often results in ancestors that are more energetically fit to the structure than their descendants, a clear contradiction.

While most sequences are associated to only one functional structure, RNA families have an old and complex history. It has been hypothesised that some ancestral RNAs were combining multiple functions, with multi-stable structures. At some point, a duplication event happened, and each copy subspecialised into a specific structure. RNA sequences are organised into evolutionary groups with conserved structure represented by stochastic context free grammars, yet some RNA sequences can align well, statistically speaking, with more than one family[2].

2 PREVIOUS WORK

To circumvent the bias introduced by reconstructing sequences when only one structure is conserved, Tremblay-Savard et al. proposed an approach using substitution and base pair costs that focuses on simultaneously reconstructing the ancestor of two related ncRNA families, assuming that they were created by this process of duplication followed by subspecialisation[3].

This approach aimed to infer ancestors that can conform to both structures, but not be as specialised to any of them as the extant sequences. A stringent simplification had to be done due to the long chain of dependencies that can exist between the two structures.



This figure shows an example of "interdependency" where paired positions (e.g. positions 3 and 5) in one structure (Fam 0 or red) are paired to different positions in the other.(Fam 1 or blue)

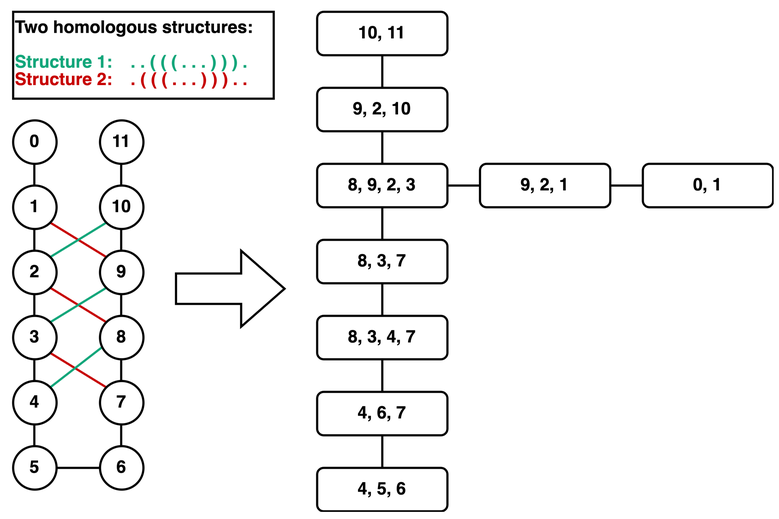
6 CONCLUSION

Results on simulated datasets demonstrate significant improvements in ancestral sequence inference accuracy while reducing the number of optimal sequences inferred by several orders of magnitude. On real datasets, we also observed that the approach is able to use more information to narrow down the space of possible ancestors. We also observed improvements at the fitness level, where the new approach is able to infer sequences that are more fit to both structures compared to previous methods.

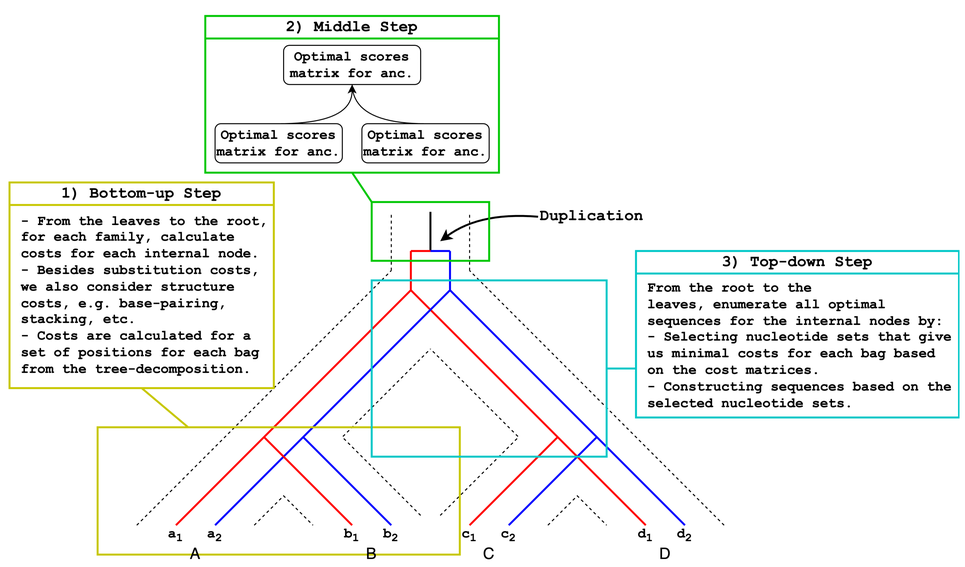
3 OBJECTIVES

In this work, we improve that previous approach (achARNement) by leveraging advances in tree decomposition algorithms to (1) incorporate simultaneously more constraints and positions in the reconstruction which (2) allows to use a more realistic energetic model.

4 METHODS



The figure shows the transformation from a **network** constructed based on base pairs, stackings, and backbone interactions to a corresponding **tree-decomposition**. (Black edges are backbones, base-pairings from structure 1 are in green, and in red for structure 2.)



In this figure, there are four species (*A, B, C* and *D*) and for each species, we have two extant RNAs. (for family 1, in *red*, and family 2, in *blue*) The three steps/stages of the algorithm are also shown.

The interdependency of the positions were simplified in achARNement. In our new approach, we applied tree-decomposition to partition the cost calculation into subproblems so that the time-complexity can be reasonably bounded.

Given a graph, a tree decomposition represents it as a tree where each node corresponds to a subset of vertices, and the tree's edges capture the relationships between these subsets. In our case, vertices in a node represent closely related positions of an ncRNA sequence (e.g. base pairs, stackings, and backbone interactions) based on its corresponding homologous ncRNA families.

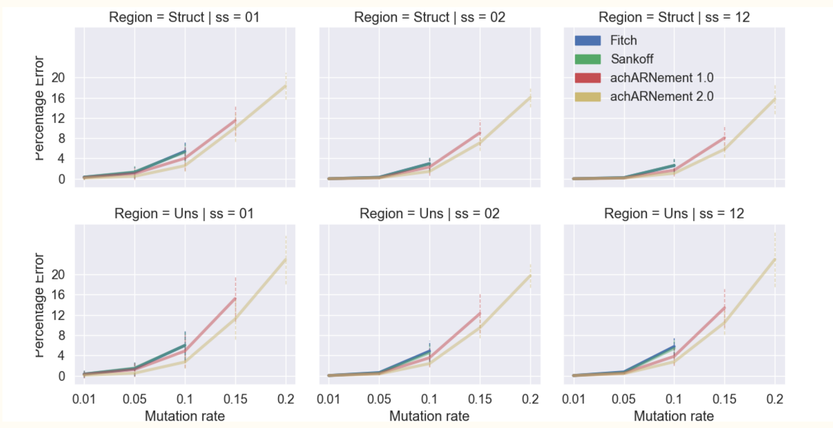
To bound the treewidth (size of the largest bag), which is a key factor for time-complexity, we applied "Tree-diet" by Marchand et al. It is a method that is able to reduce the treewidth to an alternative value while minimizing the removal of edges[4].

The algorithm consists of three steps:

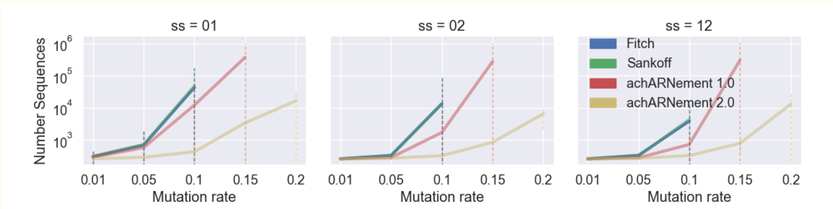
1. a bottom-up step that performs a post-order traversal to the species tree. During the process, the minimal cost which includes: a) substitution cost, and b) structure cost (minimum free energy) for each possible set of nucleotides in each bag (from the tree-decomposition) is calculated.
2. a middle step that links the minimal cost matrices for both families at the root of the phylogeny.
3. a top-down step that enumerates all the optimal sequences based on the calculated costs.

Our algorithm considers structural constraints by estimating MFE (minimum free energy), using the ViennaRNA package[5], for having specific nucleotides at given positions. This new approach allows us to reconstruct ancestral sequences that are, energy-wise, more stable than previous methods.

5 RESULTS



The average error percentage of all optimal sequences for both families in a tree. Each column represents a pair of secondary structures.



Average number of optimal sequences in the **root.** Each column represents a different pair of secondary structures with five mutation rates: 1% -> 20%.

We simulated 60 iterations of 3 homologous family pairs over 5 different mutation rates (1%, 5%, 10%, 15% and 20%). It gives us 900 trees to estimate our new approach. The results (on the left) showed an improvement in terms of accuracy over the other four algorithms. Results from Fitch, Sankoff, and achARNement are not shown for all five mutation rates. They are not able to finish enumerating the result sequences for high mutation rates.

The figure at the bottom left shows a significant improvement of our new method with regards to the total number of reconstructed sequences. The average number of inferred sequences is several orders of magnitude lower for our new algorithm. This shows that the structural constraints considered in our method have a great impact on reducing the solution space of the reconstruction.

The figure below presents results on real datasets. The average harmonic mean of energy in both structures for sequences reconstructed by achARNement 2.0 is noticeably smaller than achARNement 1.0. It means that achARNement 2.0 is able to reconstruct sequences that are more fit to both structures.

	# of Seqs		avg. Harmonic Mean of energy	
Clans	GLM clan	FinP-traJ clan	GLM clan	FinP-traJ clan
achARNement 1.0	196608	3145728	-8.6701	-10.4851
achARNement 2.0	256	384	-10.0152	-12.3875

The figure shows the results from achARNement 1.0 and achARNement 2.0 on biological data (GLM clan and FinP-traJ clan).

[1] Paten, Benedict, et al. "Genome-wide nucleotide-level mammalian ancestor reconstruction." *Genome research* 18.11 (2008): 1829-1843.
[2] Kalvari, Ioanna, et al. "Rfam 14: expanded coverage of metagenomic, viral and microRNA families." *Nucleic Acids Research* 49.D1 (2021): D192-D200.
[3] Tremblay-Savard, Olivier, Vladimir Reinharz, and Jérôme Waldispühl. "Reconstruction of ancestral RNA sequences under multiple structural constraints." *BMC genomics* 17 (2016): 175-186.
[4] Marchand, Bertrand, Yann Ponty, and Laurent Bulteau. "Tree diet: reducing the treewidth to unlock FPT algorithms in RNA bioinformatics." *Algorithms for Molecular Biology* 17.1 (2022): 8.
[5] Ronny Lorenz, Stephan H. Bernhart, Christian Höner zu Siederdissen, Hakim Tafer, Christoph Flamm, Peter F. Stadler, and Ivo L. Hofacker. ViennaRNA package 2.0. *Algorithms for Molecular Biology*, 6(1):26, 2011.

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